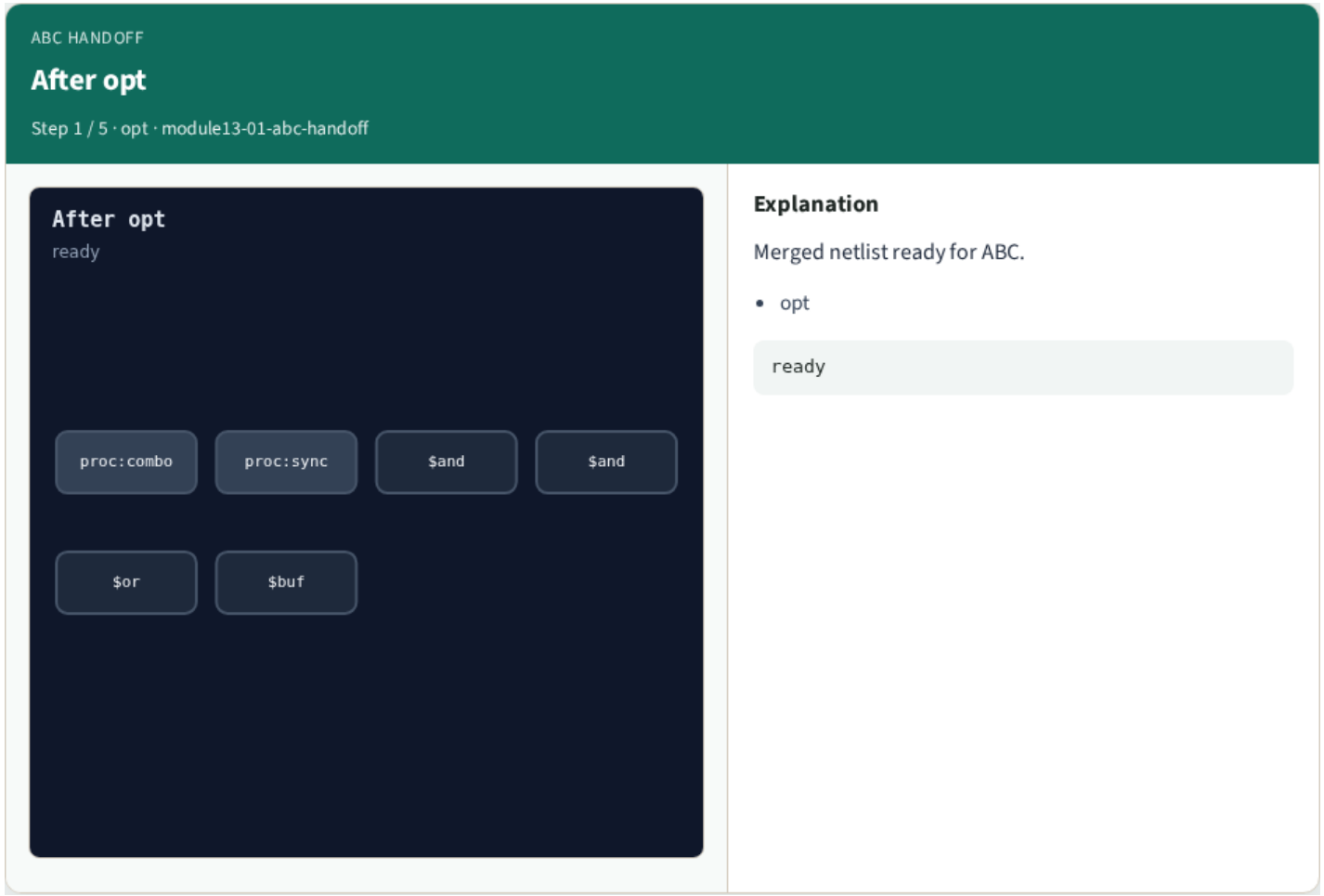
Why abc

Yosys abc is a bridge: you hand an AIG-shaped cone to an external optimizer

The idea

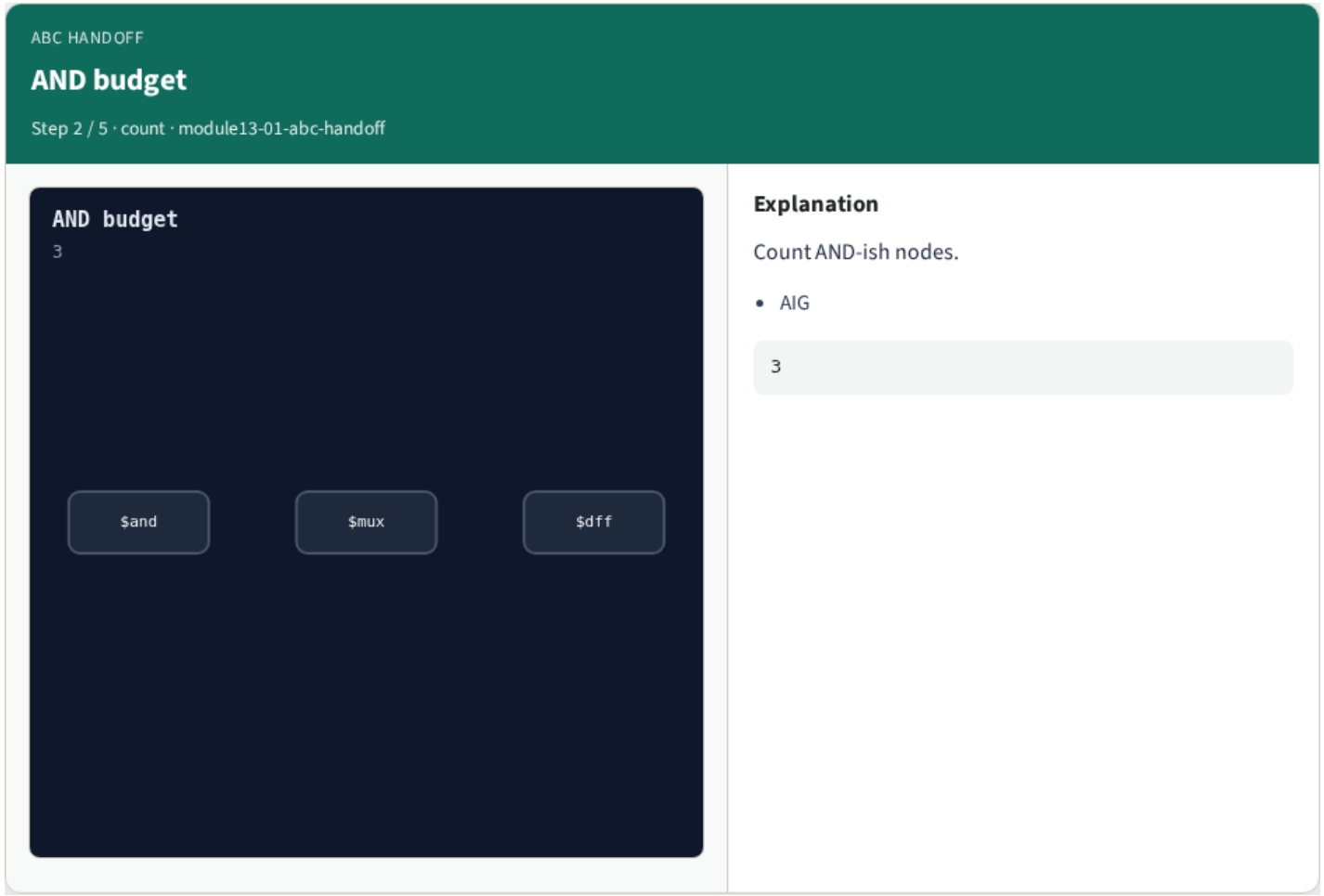
- GOLDEN: abc_aig_ands=3 after opt_merge-shaped handoff

After opt



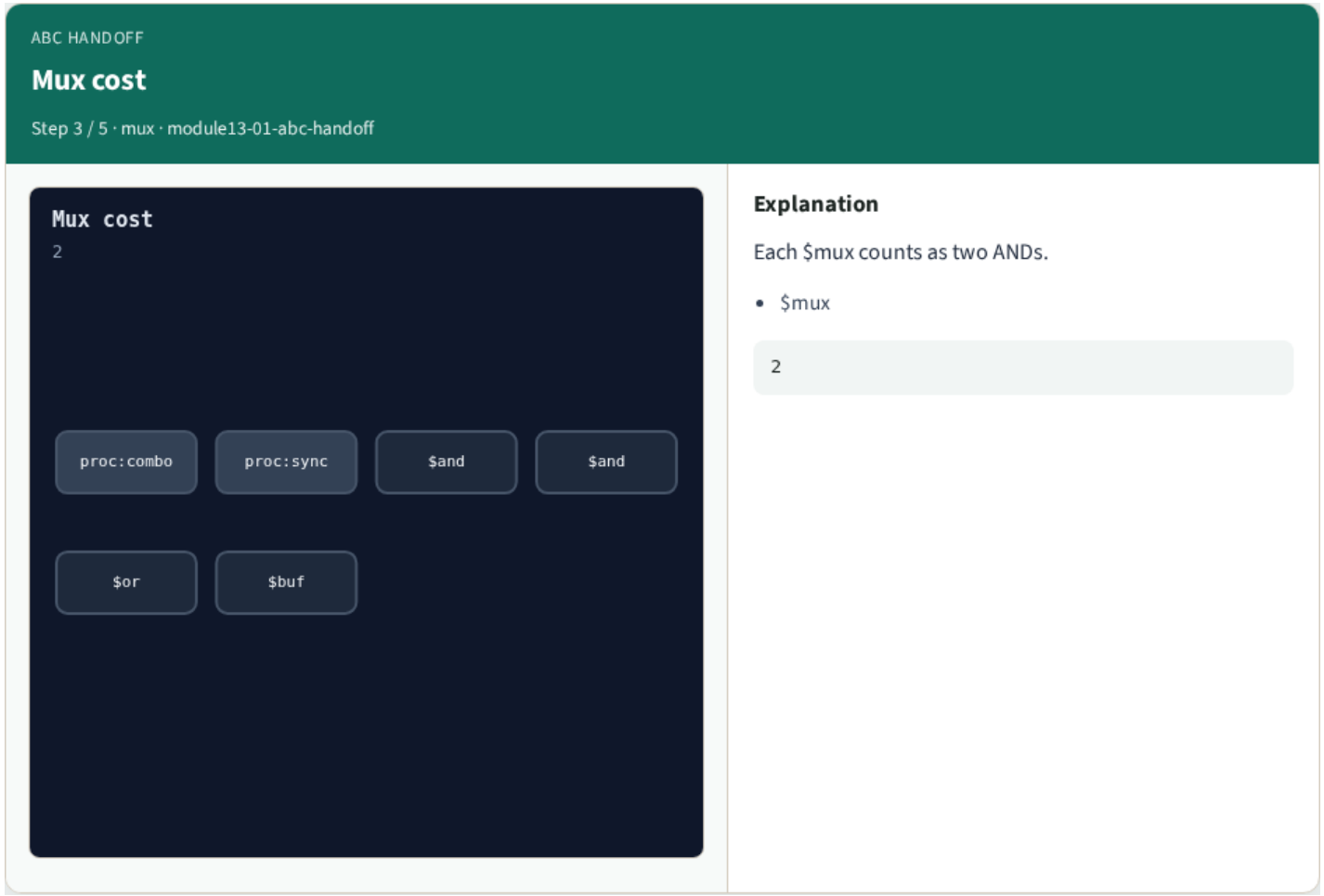
After opt

AND budget



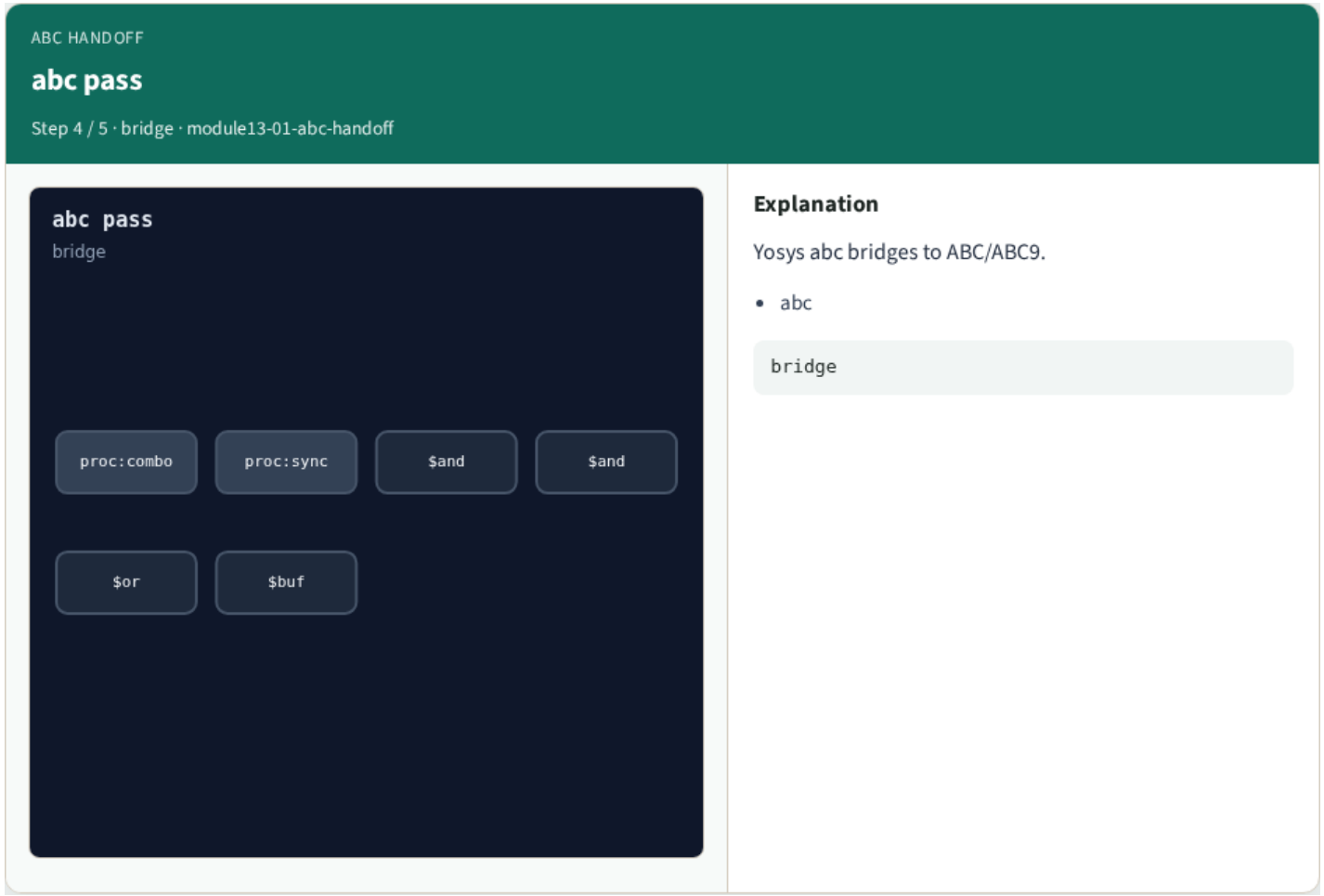
AND budget

Mux cost



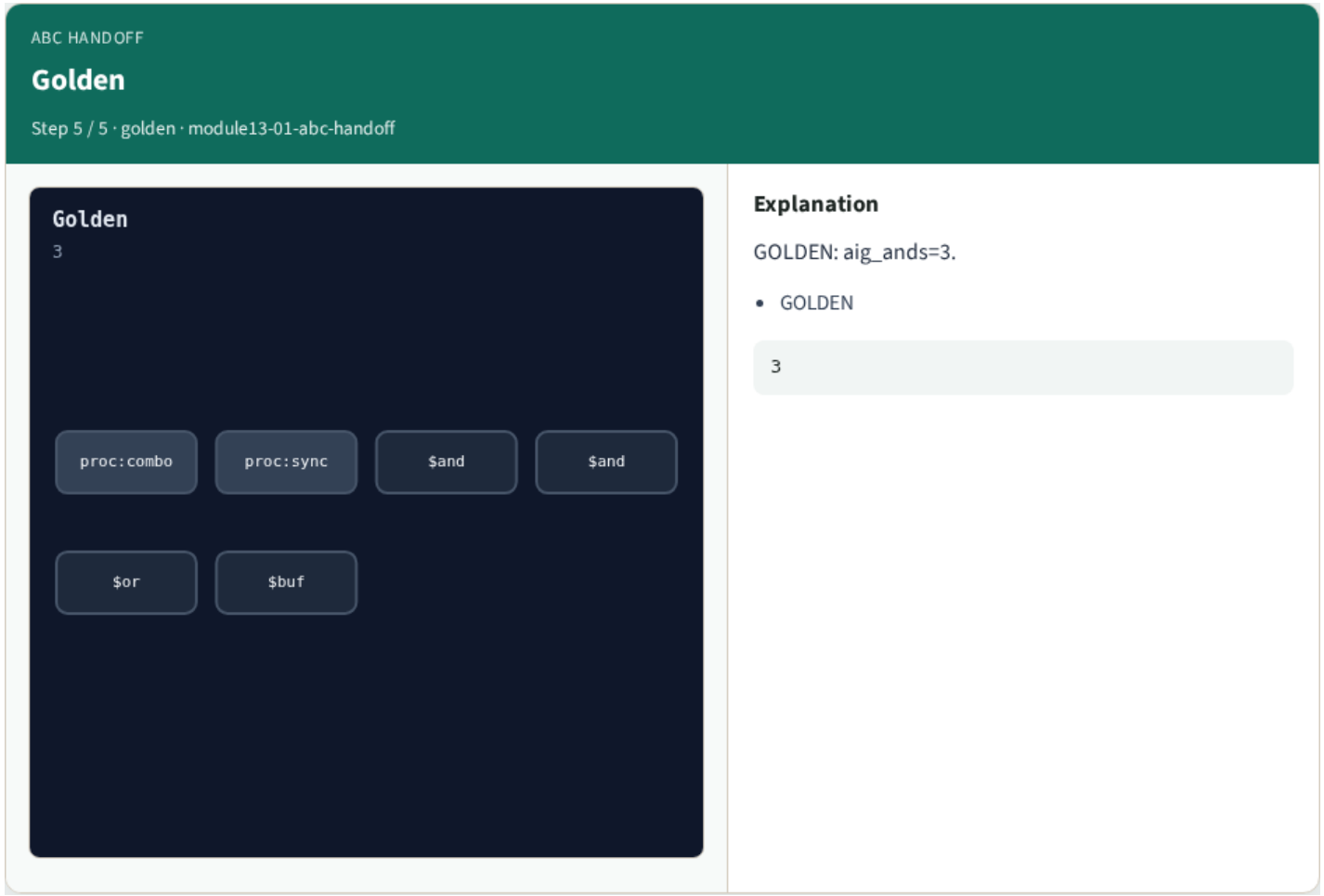
Mux cost

abc pass



abc pass

Golden



Golden

Browser lab track

- Open **abc-handoff**
- Count AND nodes for the ABC handoff

Implement track

- Call algo_abc_handoff(); assert aig_ands == 3

Pitfalls

- Counting before merge
- Treating mux as one AND instead of two in this model

Try these

- From the common folder

Try these — try these

```
cd courses/learn_yosys/common
python3 algo_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
after opt_merge
and += 1 per dollar and; += 2 per dollar mux
ok if aig_ands == 3
```

Next

- Complete the quiz for this part
- Next: full Yosys algo engine check